Murdoch University

Unit Outline

ICT303

Advanced Machine Learning and Artificial Intelligence

Teaching Period: TJA 2026

Unit coordinator: Hongen Lu

Singapore

Ngala kwop biddi.
Building a brighter
future, together.

50 YEARS

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This guide should be used in conjunction with the Handbook as the official source of information about this unit.

Refer to myMurdoch Learning for on-going communication and your learning and assessment content.

You are advised to retain this document for your records.

Acknowledgement of Country

We acknowledge that Murdoch University is situated on the lands of the Whadjuk and Binjareb Noongar people. We pay our respects to their enduring and dynamic culture and the leadership of Noongar Elders past and present. The *Boodjar* (Country) on which Murdoch University is located has, for thousands of years, been a place of learning. We at Murdoch University are proud to continue this long tradition.

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1 Unit Details

Welcome to:

ICT303

Advanced Machine Learning and Artificial Intelligence

1.1 Unit Description

This unit focuses on advanced Machine Learning (ML) and Artificial Intelligence (AI) techniques. It covers the main concepts and theories behind modern machine learning models, including the development of deep neural networks and their application to solving real-world problems. You will learn how to design, develop, implement, train, test and evaluate highly efficient deep learning models. You will also learn how to use scripting languages such as Python, which are suitable for prototyping, running and customizing existing models. Topics that will be covered in this unit include (1) linear neural networks, (2) multilayer perceptron, (3) (modern) convolutional neural networks, (4) optimisation algorithms and (5) computational performance. Students will also be introduced to various practical applications of deep learning including computer vision, natural language processing and recommender systems.

1.2 Learning Outcomes

This unit enables you to demonstrate your achievement of the following unit learning outcomes.

Unit Learning Outcomes
LO1. Explain and compare the theories and algorithms behind machine learning models
LO2. Design and implement machine learning models using appropriate programming languages.
LO3. Train, test and critically evaluate and compare the performance of machine learning models.
LO4. Apply best practices for data preparation and select appropriate data pre-processing techniques for different types of data.
LO5. Design deep learning models to solve practical problems in various domains.
LO6. Learn and use new programming and scripting languages independently, adapt to changing requirements and technologies.

1.3 Murdoch Strategic themes

As an institution, the strategic themes of Sustainability; Equity, Diversity & Inclusion; and First Nations guide and shape much of our activity (see Our Strategy 2023-2030 *Ngala Kwop Bidi*).

In this unit, we directly address the following themes in these specific ways:

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1.4 General guidance and requirements

1.4.1 Inclusivity Statement

Murdoch University strives to be a place of belonging for all staff and students. We are committed to supporting and celebrating all community members including all abilities, ethnicities or religions, sexual or gender identities. Homophobia, transphobia, racism, and ablism are not tolerated.

For help and support, please see information and contacts in Health & wellbeing in myMurdoch.

1.4.2 In case of unforeseen disruption to learning and teaching

A university or campus wide disruption may occur due to natural, political, or other human crisis (e.g., COVID-19). Where it impacts our learning, teaching, and assessment plans, then we will communicate changes relevant to this unit via myMurdoch Learning. Also pay attention to student announcements and myMurdoch for university wide information.

For individual needs, consult with Access and Inclusion to see if an individual Access and Inclusion (MyAccess) plan is appropriate, or for specific assessment item adjustments, see your Unit Coordinator.

1.4.3 Where to get help for your learning success and wellbeing

Please refer to Support & Advice via myMurdoch for all the information you need for your studies.

This includes:

- Student admin, Exams, Policies, Key dates, Complaints and appeals
- Learning and study support, including information about Academic Integrity and Murdoch Academic Passport
- Health and wellbeing information, including Accessibility services, Medical and counselling services, Aboriginal and Torres Strait Islander support, and Sexuality and gender diversity support
    - Contact Access and Inclusion if you have a disability, Specific Learning Disorder or health condition (including mental health condition), are pregnant or breastfeeding or are caring for a person with a disability, **and** require support in accessing your studies.
    - Kulbardi Aboriginal Centre provides support for Aboriginal and Torres Strait Islander students.
- myMurdoch Career offers resources to help you use your current learning to prepare for your future career.

1.4.4 Key dates and considerations for withdrawing from this unit

If you are considering withdrawing from this unit, see the Withdrawing page for general information and implications.

See the Teaching Periods page for implications of withdrawing at different times of the teaching period, including Census Date, and search for the specific dates for your current teaching period.

1.4.5 Where to find your class

To find any location at Murdoch, use Murdoch Maps.

2 Contact details

2.1 Unit coordinator

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Name:	Hongen Lu	
Preferred name and pronouns:	Name:	Pronouns:
School:	School of Information Technology	
Campus:	Murdoch	
Email:	Hongen.Lu@murdoch.edu.au	
Phone:		

2.2 Teaching team

You will be notified who your teaching team is at the beginning of the teaching period. The teaching team member will provide you with their contact details.

Name:	TBA	
Preferred name and pronouns:	Name:	Pronouns:
Role:		
Campus:		
Email:		
Phone:		

3 Overall design and expectations

3.1 Approach to learning and teaching

The focus of this unit is on advanced machine learning. This involves a deep understanding of the mathematical concepts behind modern machine learning models, their implementation details and their efficient usage. You will also learn how to design machine learning models to solve specific problems.

The weekly lectures cover a number of important theoretical concepts of machine learning and demonstrate their applications and importance via concrete examples. Prior to attending their lab session each week, students are strongly advised to

- attend the lectures in order to develop an understanding of these examinable concepts and
- review all relevant theoretical material (primarily from the lecture notes available on LMS, but also readings)

In **the weekly labs,** which will focus on the practical aspects. The lab activities will involve programming neural networks and deep learning models using appropriate programming languages. You will work through the lab exercises individually, guided by the teaching staff. **It is important that you come prepared for the lab by reading the lecture materials and additional readings.** If you are not prepared for the tutorial and lab sessions, you will not gain as much of the benefit from completing it as you should.

It is strongly advised to review the lab questions in advance and attempt to prepare either a draft of a solution or otherwise make notes to assist you in solving the problem. **It is a very bad practice and a bad learning habit to come to a lab session unprepared, i.e., without reading the lecture materials.**

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Lab sessions are generally conducted in an interactive manner, with considerable dialogue between students and tutor, both individually, small groups and as a class. Laboratory sessions are intended as learning exercises first, and therefore constitute formative assessment.

Students who miss even a small number of lectures and lab sessions (one or two) can quickly get behind and may find it difficult to catch up. **If you miss a lecture session**, you are strongly advised to read the lecture materials yourself, watch the pre-recorded videos (if any) and watch the recorded live lecture. Due to the amount of interactive material used in lectures, lecture recordings may not capture all of the information conveyed in lectures. Students who are unable to regularly attend lectures should anticipate spending an additional 2 to 4 hours of revision per week.

If you miss a lab session, you are strongly advised to either attempt to attend a different session (with permission of the unit coordinator or tutor), or otherwise to make up this work independently in your own time.

3.2 Expected time commitment for studying this unit

As this is a 3credit point unit, we expect you to spend on average 10.7hours per week for the total weeks of this teaching period (or 150 hours overall) working on this unit.

3.3 Unit changes in response to student feedback

Previous students have provided us with feedback to improve this unit. Changes we have incorporated this time:

-

Please help us continue this feedback cycle by completing your own unit survey which will open toward the end of teaching. You can find your Myfeedback surveys at http://myfeedback.murdoch.edu.au/.

4 Unit Schedule

See myMurdoch Learning for details of all learning activities and assessments. Go to TeachingTimetables or myCalendar to see your scheduled class times.

Timetable specifics subject to change. You will be notified of changes by the Teaching Team.

Week	Topic	Learning activity	Assessment due date
1	Introduction and Overview	Lecture and Tutorial	
2	Linear Neural Networks	Lecture and Tutorial	
3	Multilayer Perceptron	Lecture and Tutorial	
4	Optimization and backpropagation algorithm	Lecture and Tutorial	
5	Convolutional Neural Networks Part I	Lecture and Tutorial	
6	Convolutional Neural Networks Part II (we will also cover training)	Lecture and Tutorial	
7	Recurrent Networks (RNN, LSTM, etc.)	Lecture and Tutorial	Assignment Due Friday 23:59 of Study Week 7

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Week	Topic	Learning activity	Assessment due date
8	Applications – Part I (Computer Vision, Natural Language Processing, Recommender Systems)	Lecture and Tutorial	
9	Applications – Part II (Computer Vision, Natural Language Processing, Recommender Systems)	Lecture and Tutorial	
10	Generative AI	Lecture and Tutorial	
11	Large Language Models (LLMs)	Lecture and Tutorial	
12	Unit Review and Preparation for the Final Exam	Lecture and Tutorial	Assignment two due Friday 23:59 of Study Week 12

5 Assessments

Assessment for this unit is conducted in accordance with the Assessment Procedure and Student Assessment Support Procedure.

Refer to myMurdoch Learning for full Assessment details, including additional requirements.

5.1 Assessment summary

No	Assessment Name	Unit Learning Outcomes	Weight %	Individual / Group	Data of release	Due Date and Time
1	Assignment 1	LO1, LO2, LO3, LO6	25%	Individual	Monday 8:00 AM of Study Week 5	Friday 23:59 of Study Week 7
2	Assignment 2	LO1, LO2, LO3, LO4, LO5, LO6	25%	Individual	Monday 8:00 AM of Study Week 11	Friday 23:59 of Study Week 12
3	Final Examination	LO1, LO2, LO3, LO4, LO5	50%	Individual	Specified exam period	Specified exam period

5.2 Assessment information

5.2.1 Assessment 1

ASSESSMENT OVERVIEW

[OFFICIAL]

This is a take-home assignment, which should be completed within 4 weeks of its release. It is composed of one or multiple exercises in which you will be required to design a machine learning model, implement it using an appropriate programming language, test it and evaluate its performance.

This is an individual work and thus must be completed individually.

WHERE TO SUBMIT

HOW IT IS ASSESSED (summary)

Submission details are given in the section “Assignment 1” in LMS. Please pay very close attention to all of these instructions as assignments which do not meet these criteria may lose marks or not be accepted.

Note that exact due dates and times are available on LMS.

Uploaded submissions should be made as a single zip files:

- Copy all your files into a folder named ICT303_StudentID_Surname_Assignment1. For instance, student John Smith whose student ID is 123456, should name the folder as: ICT303_123456_Smith_Assignment1
- Zip the folder using winzip (RAR files will not be accepted). This will result in a file named ICT303_StudentID_Surname_Assignment1.zip.
- Upload this file to the submission area on LMS.

Note that any piece of work or assessable component may not be accepted or otherwise awarded zero marks if not submitted in an appropriate form as per supplied instructions.

The assessment includes a moderation process to ensure reliable, just, and fair outcomes.

Your assessment outcome will be provided to you LMS.

- If you are an internal student, you can discuss your feedback face-to-face with your tutor.
- If you are an external student, you can also request to discuss your feedback with the tutor who is assigned to handle external students.

ESSENTIAL ADDITIONAL INFORMATION

See myMurdoch Learning for further details, such as instructions, communication, resources, guides, exemplars and a descriptive rubric assessment tool.

5.2.2 Assessment 2

ASSESSMENT OVERVIEW

This is a take-home assignment, which you are required to complete within 4 weeks of its release. It is composed of one or multiple exercises in which you will be required to design a machine learning model, implement it using an appropriate programming language, test it and evaluate its performance. The problem that needs to be solved will be more complex.

This is an individual work and thus must be completed individually.

WHERE TO SUBMIT

[OFFICIAL]

Submission details are given in the section “Assignment 2” in LMS. Please pay very close attention to all of these instructions as assignments which do not meet these criteria may lose marks or not be accepted.

Note that exact due dates and times are available on LMS.

HOW IT IS ASSESSED (summary)

Uploaded submissions should be made as a single zip files:

- Copy all your files into a folder named ICT303_StudentID_Surname_ Assignment2. For instance, student John Smith whose student ID is 123456, should name the folder as: ICT303_123456_Smith_Assignment2
- Zip the folder using winzip (RAR files will not be accepted). This will result in a file named ICT303_StudentID_Surname_ Assignment2.zip.
- Upload this file to the submission area on LMS.

Note that any piece of work or assessable component may not be accepted or otherwise awarded zero marks if not submitted in an appropriate form as per supplied instructions.

The assessment includes a moderation process to ensure reliable, just, and fair outcomes.

Your assessment outcome will be provided to you via LMS.

5.2.3 Final Exam

ASSESSMENT OVERVIEW

The final exam will occur in the assessment period after the end of semester. Please ensure that you are aware of all University requirements with regards to final exams. The examination is timetabled and run by the University Examinations Office. The exam timetable is usually released in the second half of the semester.

The examination will run for 2 hours and be closed book. The structure of the examination will typically be short answer questions including those requiring the writing of algorithms and/or program code in the language studied during the semester.

Please be aware of the University's requirements regarding what may be brought into a closed book examination and the identification acceptable when attending an exam. For further information about examinations, refer to http://www.murdoch.edu.au/oss/exams/

This is an individual work and thus must be completed individually.

The exam will be **face-to-face and supervised.**

Note that any piece of work or assessable component may not be accepted or otherwise awarded zero marks if not submitted in an appropriate form as per supplied instructions.

5.3 Academic integrity

Murdoch University expects students and staff to pursue the highest standards of integrity in all academic activity. Academic integrity involves behaving ethically and honestly in all learning and teaching activity and relies on respect for others’ ideas through proper acknowledgement and referencing of publications.

More information about academic integrity can be found at https://goto.murdoch.edu.au/AcademicIntegrity. To help you learn about academic integrity practices, all students are required to complete the Murdoch Academic Passport (MAP100). Please also note the library citation guide.

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5.3.1 Use of Artificial Intelligence including generative AI

Murdoch University supports the embedding digital literacy and emerging technologies (GenAI) as a learning expectation (where appropriate) in recognition of its increasing role in society.

You must engage in appropriate and ethical use of Generative Artificial Intelligence (GenAI) and other forms of AI use. This includes acknowledging, citing and referencing use in all your work.

You must follow the directions of use for this unit as outlined in this Unit Outline, and in myMurdoch Learning.

Any suspected inappropriate use will be investigated for misconduct.

5.3.2 Misconduct

Academic misconduct is treated seriously and penalties may apply.

Your Unit Coordinator may apply a range of approaches to verify that your submitted assessment is your own work.

Text Matching Software

Murdoch University makes use of text matching software and other forms of comparison to identify unoriginal, incorrectly referenced, or uncited text. Where possible, assessments will be submitted to Turnitin.com for the detection of plagiarism and the unauthorised use of artificial intelligence. All submitted assessments will be retained as source documents in the Turnitin.com reference database. You will be required to agree to the Turnitin End-User License Agreement when submitting.

Other methods

Other methods and tools may be used to confirm academic integrity, and these may include reviewing ownership and progress steps.

5.4 Extensions and late submissions

This unit follows Murdoch policies and procedures, particularly the Student Assessment Support Procedure and Assessment Procedure, with regards to assessment submission and extensions, supplementary and deferred assessment, and other expectations.

In this unit, extensions and late submissions follow these requirements:

- Assessments submitted after the due date without approval will not be marked, and you will receive 0% for the assessment.
- You can apply for an assessment extension without penalty. See the Student Assessment Support Procedure for all requirements.

Students who feel that their disability, health condition or disability caring responsibilities may impact on their capacity to meet assessment submission are strongly advised to visit Access and Inclusion as early as possible to discuss potential needs and assistance.

5.5 Determination of the final grade

Refer to the Grades Policy for information about marks and grades.

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6 Learning resources

Your learning resources and any updates are provided through myMurdoch Learning (LMS), including My Unit Readings.

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7 Academic Advice and Student Support

Need guidance on study related issues?
Use this flowchart or seek direct assistance from Student Support Services or MyMurdochAdvice.

Contact your Tutor if you have . . .

- Questions about content covered in tutorials or practical sessions.
- General questions about completing assessments.
- Concerns about another student or your learning needs.
- Positive and constructive feedback.

→ Tutor

Contact your Unit Coordinator if you have . . .

- Questions about unit content, assessments, attendance or tutorial times.
- Questions on marked assessments.
- Request a re-mark/review of a marked assessment.
- Academic issues with your learning in this unit.
- Positive and constructive feedback.

→ Unit Coordinator

Contact your Academic Chair if you have . . .

- Academic and assessment issues that haven't been adequately addressed by the Unit Coordinator.
- Academic issues relating to progression through your degree, withdrawal from a unit or intermission.
- Positive and constructive feedback.

→ Academic Chair

Contact your Head of School if you have . . .

- Academic, assessment or other issues that haven't been adequately addressed by your academic chair or you aren't comfortable discussing with your Academic Chair.
- Formal request for re-mark/review in accordance with policy and procedure, where Unit Coordinator has not undertaken.
- Complaints or appeals relating to your studies that haven't been adequately addressed. Visit Complaints and Appeals for more advice.
- Positive and constructive feedback.

→ Head of School

To further escalate an appeal or complaint, contact the Associate Dean Learning and Teaching and/or see Complaints and Appeals for formal appeals procedures.

STUDENT SUPPORT SERVICES: https://goto.murdoch.edu.au/supportservices

LEARNING AND STUDY SUPPORT: https://goto.murdoch.edu.au/learningstudy

MYMURDOCH ADVICE: https://goto.murdoch.edu.au/mymurdochadvice

COMPLAINTS AND APPEALS: https://goto.murdoch.edu.au/ComplaintsAppeals

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